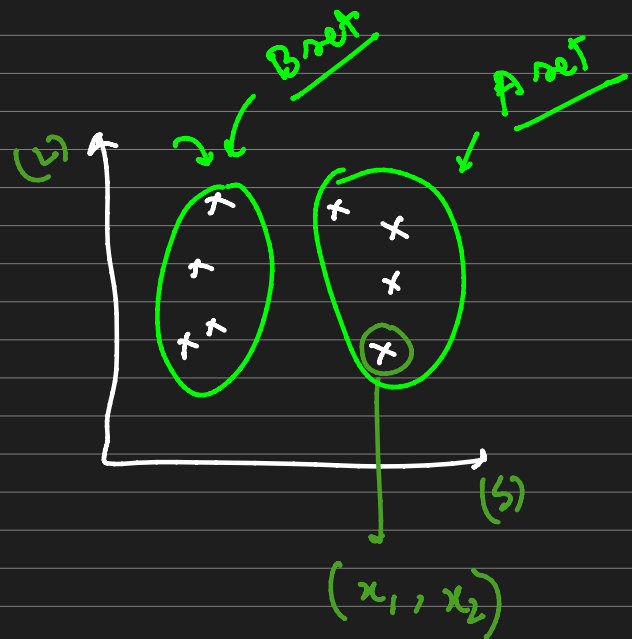


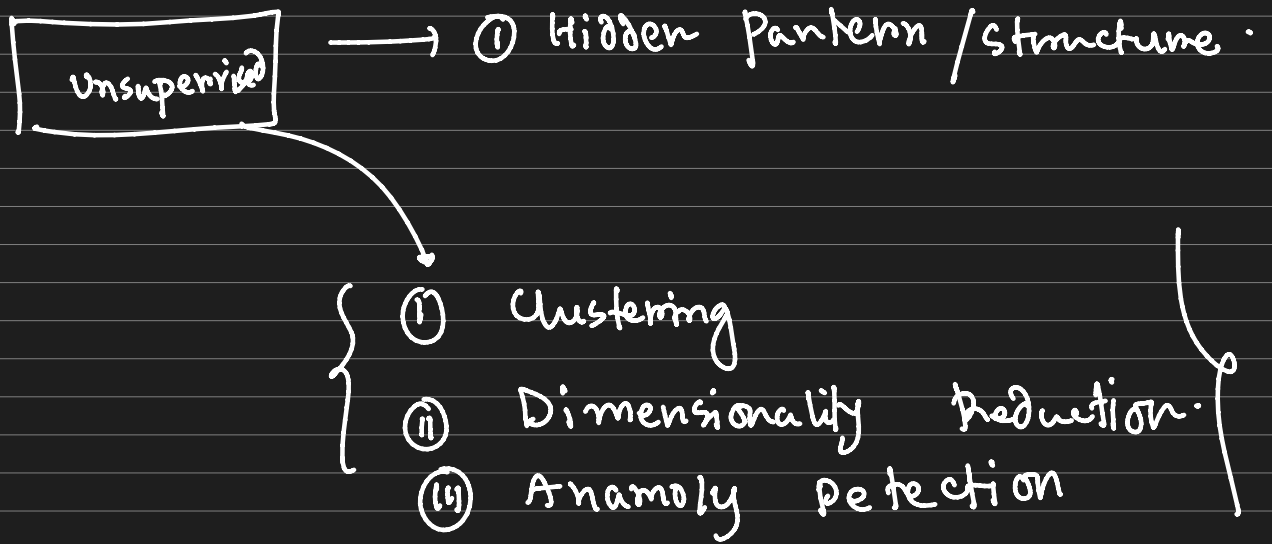
Dataset Diff:

<u>S</u>	<u>L</u>	<u>A</u>
4	5	F
7	2	F
3	8	F

model

S	L
4	5
7	2
3	8

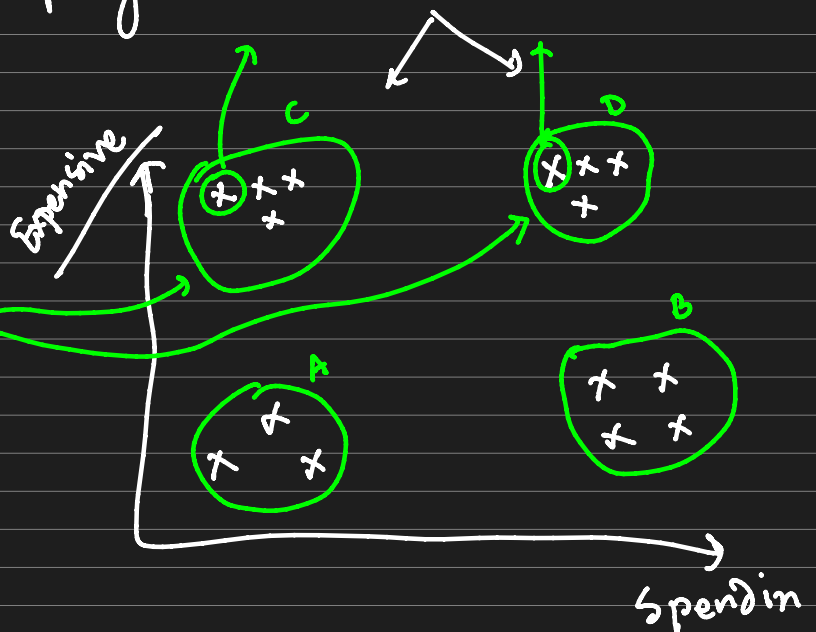
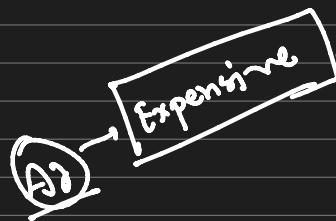


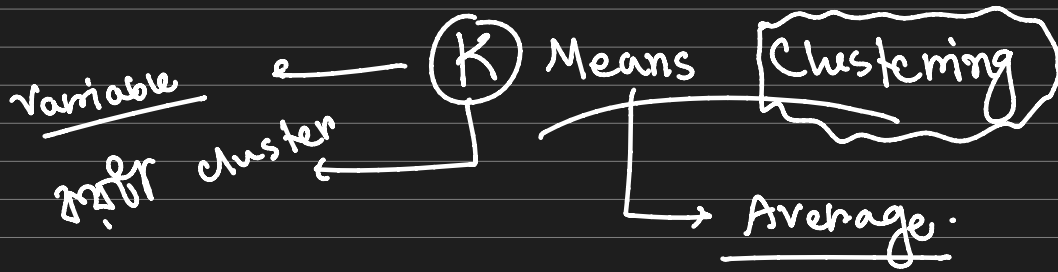


Clustering → Grouping

Ex:

E-commerce





Procedure:

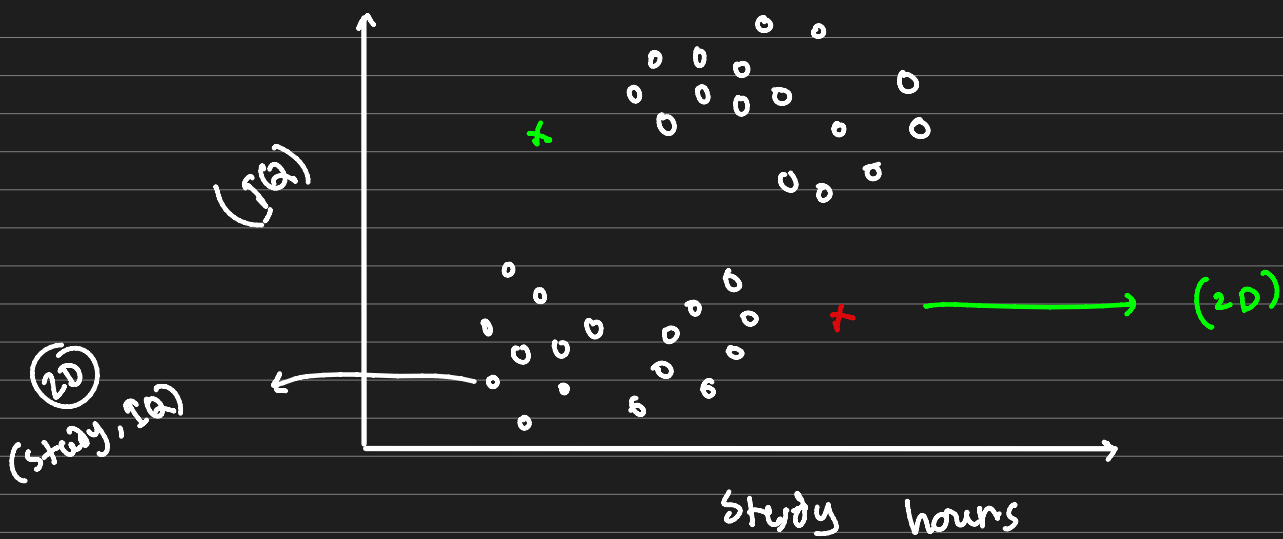
① K cluster? $\rightarrow$ K is 2 or 3



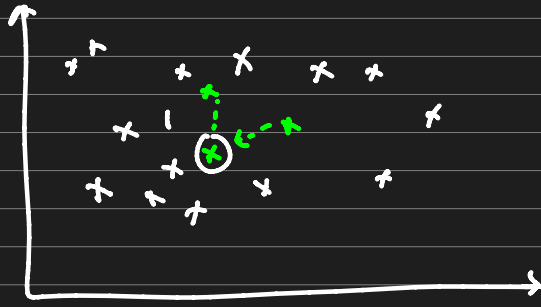
st	IQ
-	-

$K=2$

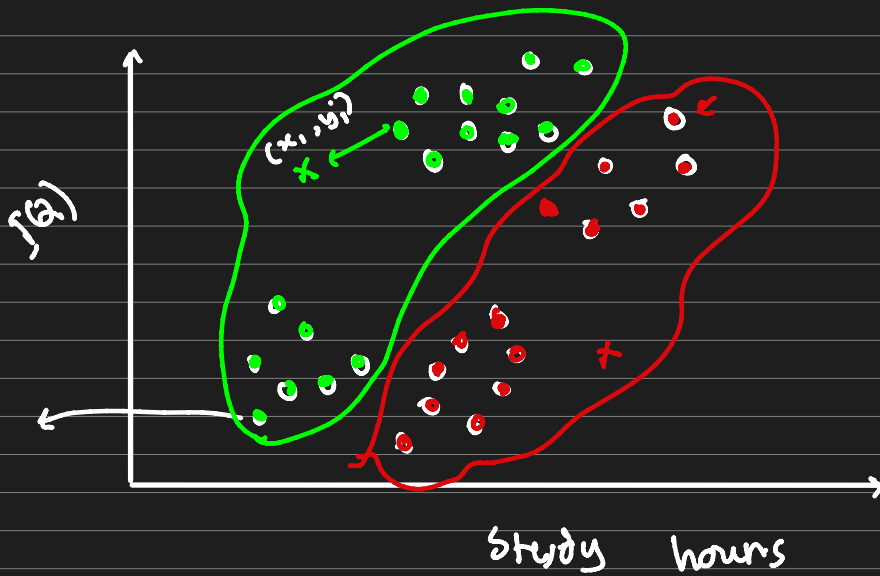
Or, random points & diff centroid create 2 clusters.



Centroid: $\rightarrow$ $\frac{\sum x_i}{n}$.



2nd: cluster 2 points assign via distance measure $\frac{\sum (x_i - \mu)^2}{n}$

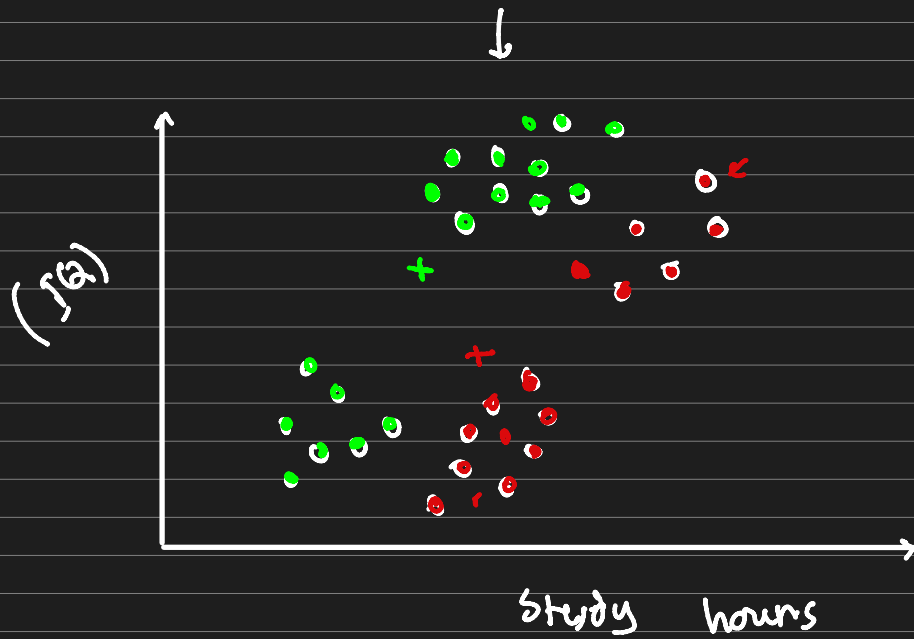


$$(2)^n = 4$$

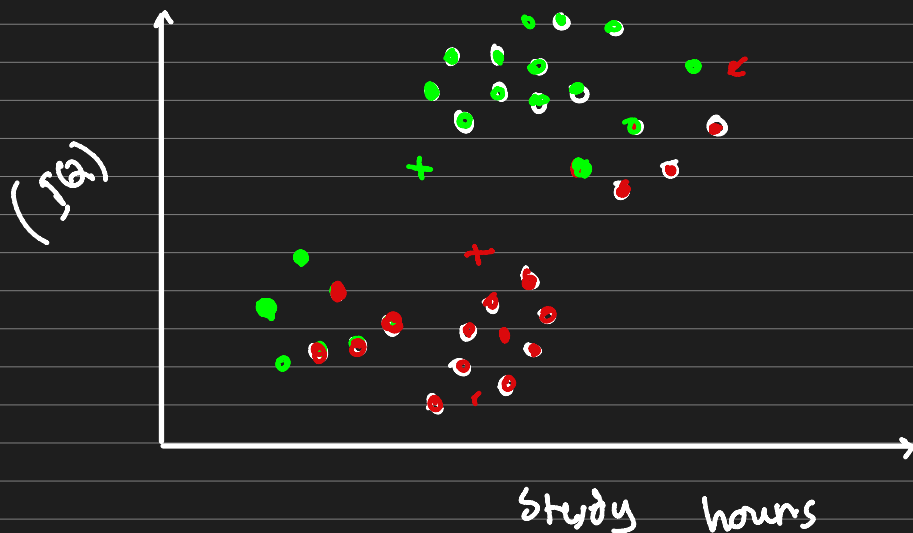
Distance:

$$\left\{ \|x_2 - x_1\| + \|y_2 - y_1\| \right\}^2$$

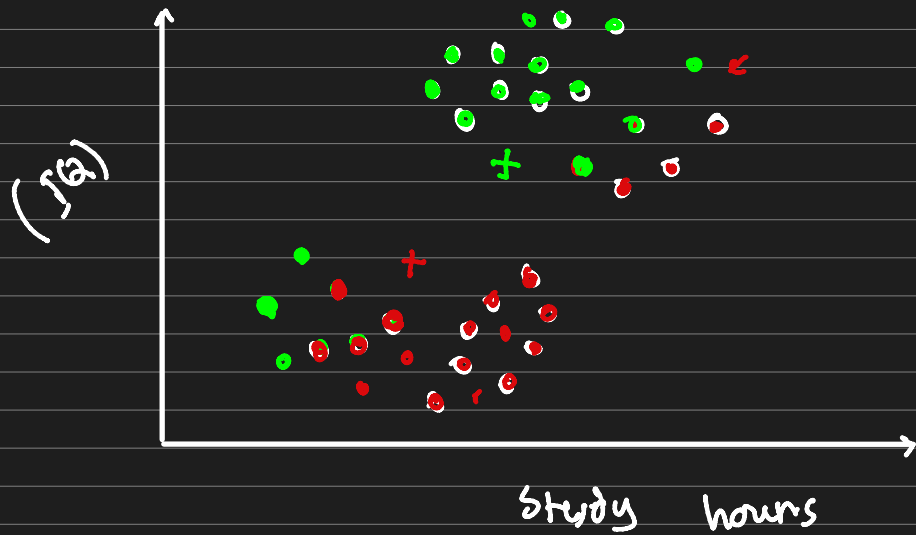
① Centroid shift update and, correct clustering for



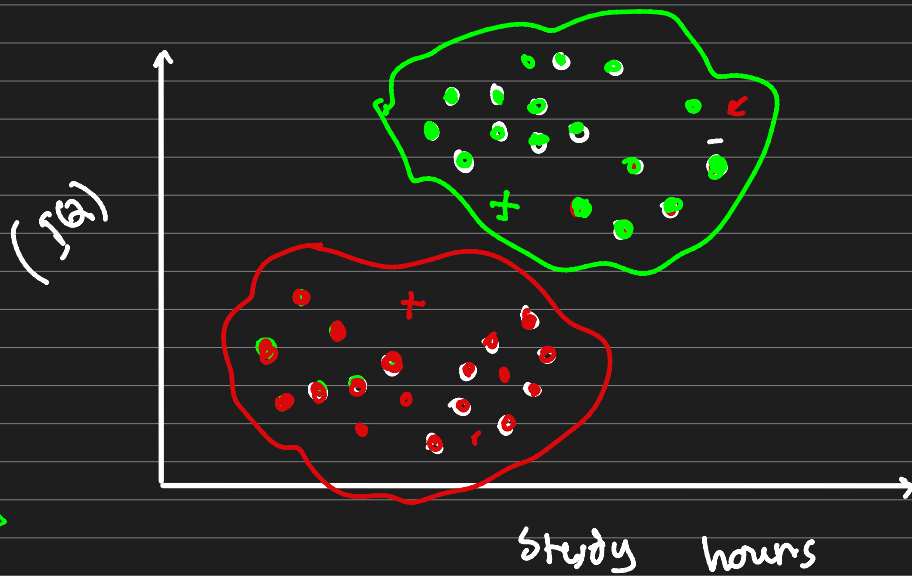
② Data-points newer centroid and base
and re clustering



① 2nd time centroid replacement

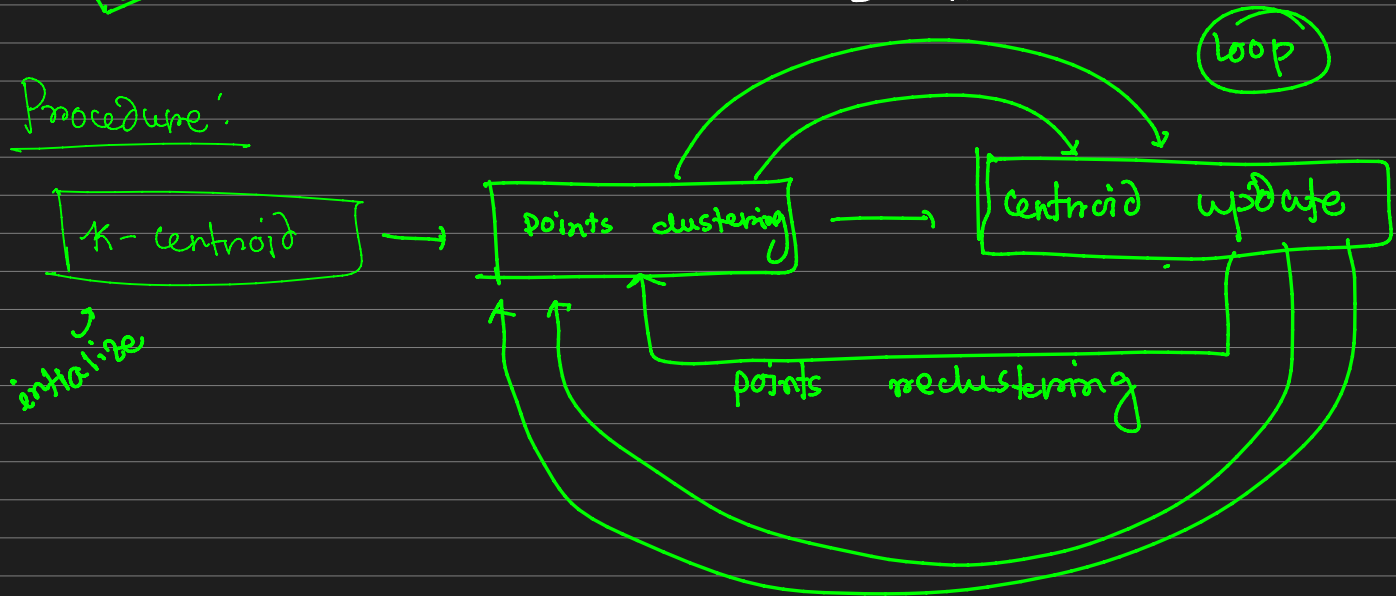


④ points reassigning



finite

Procedure:

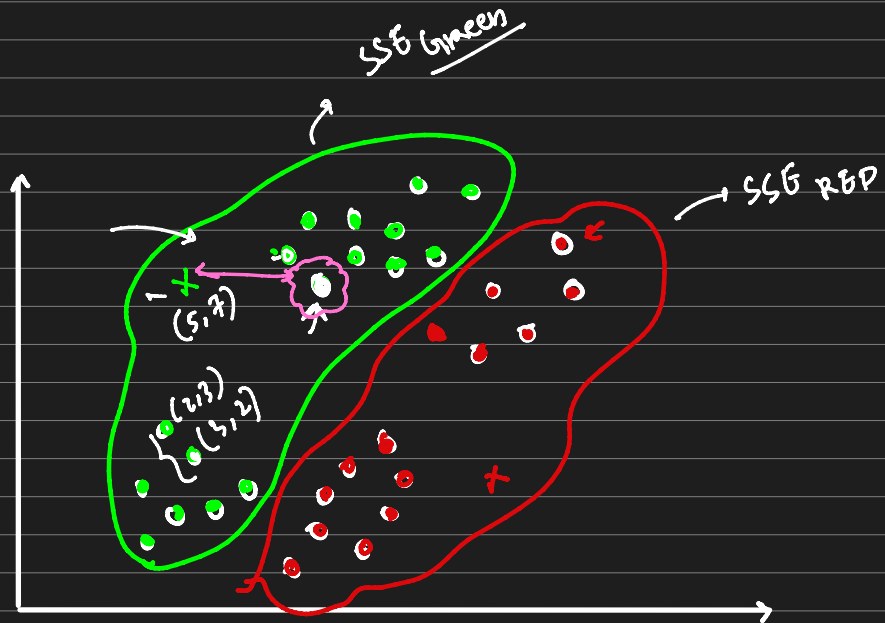


Cost function / Objective function

SSE (Sum of Squared Error)

$K=2$

data



Sum of Squared :

Cluster 1 → Green

① G₁-cluster 20 centroid (2, 3)

② cluster 20 centroid (7, 3)

Ex:

$$A = (5-3)^2 + (7-2)^2$$

$$B = (5-2)^2 + (7-3)^2$$

$$SSE = A + B$$

cost of error

centroid of error

point.

Cluster 2 → Red

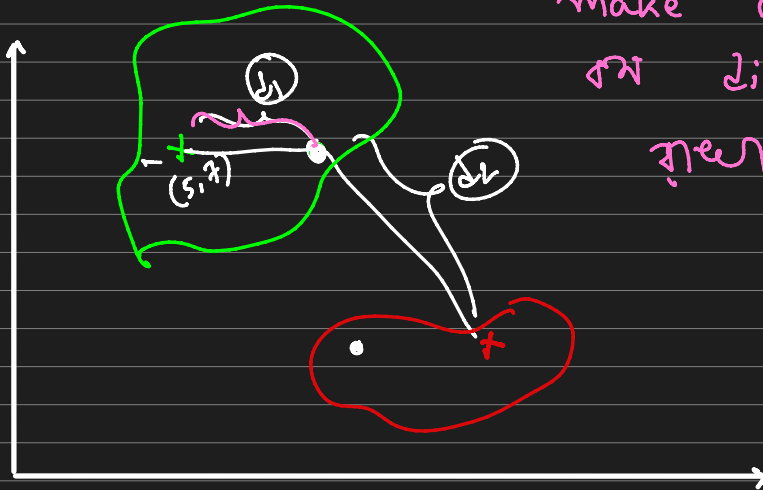
$$\therefore \text{Total SSE} = SSE_{RED} + SSE_{Green}$$

Why cost always decrease on stay constant.

① points clustering → centroid fixed

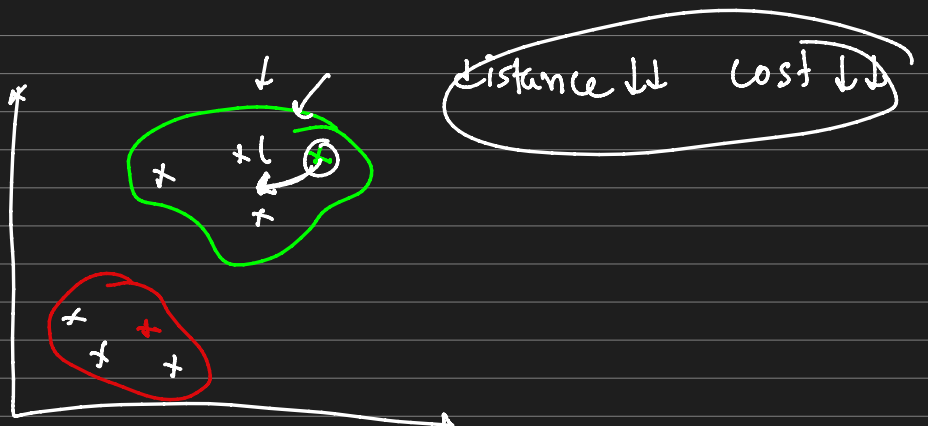
↳ cost (not add 2nd → 4th)

make more for 2nd.
for distance of centroid
near

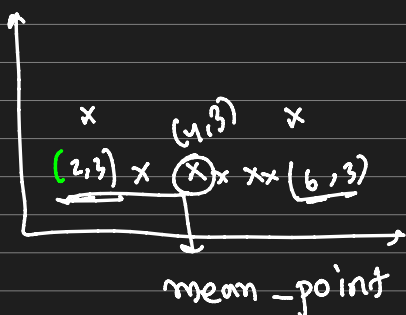


② centroid update

↳ points same cluster 2nd.



distance ↓ cost ↓



$$\begin{cases} (3-2)^2 + (3-3)^2 = 1^2 + 0^2 \\ (6-3)^2 + (3-3)^2 = 3^2 \end{cases}$$

$$1^2 + 3^2 = 1 + 9 = 10$$

$$\frac{6+2}{2} = (4, 3)$$

$$\text{dis} = (4-2)^2 + (6-4)^2$$

$$= 2^2 + 2^2 = 4 + 4 = 8$$

let's code K Means Clustering.

④ $n_clusters$: Number of Cluster

- ① Domain knowledge →
- ② Elbow → {cost}

